

Machine Learning Approach for Telco Churn Prediction



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Introduction

In the highly competitive telecommunications industry, retaining customers is far more cost-effective than acquiring new ones. This project uses machine learning to analyze customer behavior and identify churn risks early. The analysis is based on the Telco Customer Churn dataset from Kaggle, which includes customer demographics, account details, subscribed services, and churn status. With accurate churn predictions, companies can design targeted retention strategies to reduce revenue loss and support long-term business sustainability.

Data Understanding

7043 Rows & 21 Cols

→ 1 Identifier Variable

→ 3 Numerical Variables

→ 18 Categorical Variables

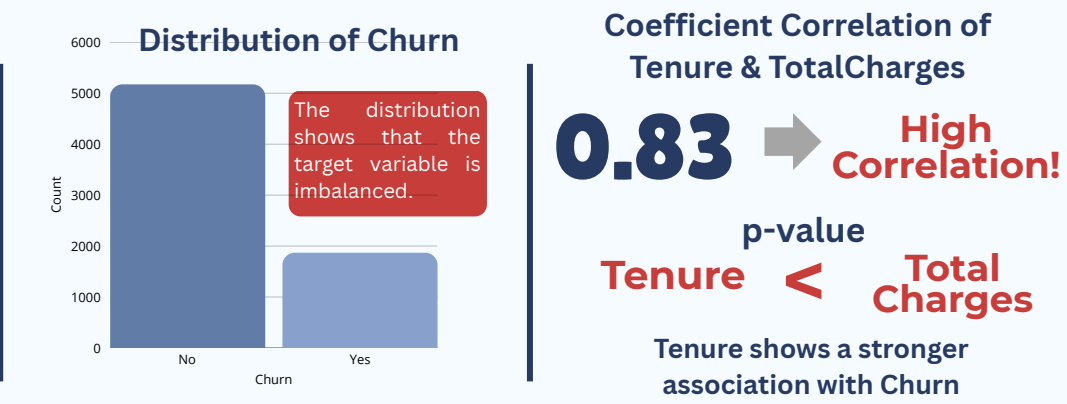
→ 1 Target Variable (Churn)

Data Problems

★ Missing Values

★ Inconsistent Format

★ Inappropriate Data Type



Feature Selection

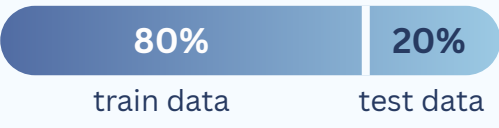
- ✗ CustomerID

no significant impact
- ✗ gender

no significant impact
- ✗ TotalCharges

avoid multicollinearity

Splitting Data



Data Preprocessing

- Numerical Missing Values

are handled by median imputation due to skewed data distribution.
- Categorical Missing Values

are handled by mode imputation.
- Inconsistent Format

Each category should follow the same writing format.
- Inappropriate Data Type

Typecasted TotalCharges column into float data type.

Feature Engineering

- Encoding

Label Encoder → binary categorical features

One-Hot Encoder → nominal categorical features
- Scaling

Robust Scaler → MonthlyCharges

Robust Scaler was selected due to non normal distribution.
- PCA

99% Variance captured with 2 components.

Modeling & Evaluation

Models trained:

• Logistic Regression

• CatBoost

• SVC

• XGBoost

• Decision Tree

• LightGBM

• Naive Bayes

• AdaBoost

• KNN

• Gradient Boosting

• Random Forest

Due to the imbalanced distribution of the target variable, either **SMOTE** or **Class Weighting Parameter** is used to better represent the minority class.

Performance Metric

Due to class imbalance, the evaluation focusses on **Precision Class 0** to avoid mislabeling non-churn customers and **Recall Class 1** to capture as many churn customers as possible.

	SMOTE		Class Weighting Parameter	
	Precision 0	Recall 1	Precision 0	Recall 1
Logistic Regression	0.895	0.778	0.894	0.778
SVC	0.895	0.789	0.895	0.789
Decision Tree	0.832	0.570	0.815	0.487
Naive Bayes	0.904	0.824	0.000	0.000
KNN	0.859	0.655	0.000	0.000
Random Forest	0.853	0.626	0.822	0.471
CatBoost	0.887	0.733	0.000	0.000
XGBoost	0.882	0.719	0.870	0.682
LightGBM	0.890	0.743	0.888	0.738
AdaBoost	0.898	0.781	0.811	0.465
Gradient Boosting	0.900	0.767	0.895	0.759

Top Models

SMOTE showed greater impact due to its ability to generate synthetic minority data.

Model	Precision 0	Recall 1
Logistic Regression	0.92	0.86
SVC	0.89	0.77
Naive Bayes	0.90	0.82
AdaBoost	0.90	0.81

Hyperparameter tuning had minimal impact on SVC, Naive Bayes, and AdaBoost, while **Logistic Regression** showed the **most improvement**, especially in recall, making it the **best model for churn prediction** in telco customer dataset

Conclusion

SMOTE achieved better performance metrics than Class Weighting Parameter by more effectively addressing class imbalance. With this improvement, Logistic Regression emerged as the best-performing model after model comparison and hyperparameter tuning, particularly in improving recall, which is crucial for identifying churn customers. With strong regularization, Logistic Regression remains simple, robust, and generalizable, making it a reliable approach for churn prediction in this dataset.

The full code is available on Google Colab.
Scan this QR to access full code.